

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 10
Orthonormal Set and Gram-Schmidt Algorithm.

Review

1. Show: if $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$ are linearly independent and $\vec{x}$ is a linear combination of them (so $\vec{x} = \beta_1 \vec{a}_1 + \dots + \beta_k \vec{a}_k$) then the coefficients $\beta_1, \dots, \beta_k$ are unique.
2. What is a *basis* (of $\mathbb{R}^n$)?
3. Check that $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$ are linearly independent (in $\mathbb{R}^n$).
4. What is the linear-dependence inequality?
5. Why is the linear-dependence inequality true?

New (Chp. 5)

1. What does it mean for a set of vectors to be *orthogonal*?
2. What does it mean for a set of vectors to be *orthonormal*?
3. Check that $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$ are an orthonormal set (in $\mathbb{R}^n$).

4. Why are orthonormal vectors linearly independent?

5. What is the *orthonormal expansion* of $\vec{x} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$ with respect to the orthonormal list $\vec{e}_1, \vec{e}_2, \vec{e}_3$?

6. Apply the Gram-Schmidt algorithm to the input set $\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \vec{a}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Are the (1) input vectors linearly dependent, or (2) the output vectors are orthonormal? Check.